Assignment 3: Time-Series Weather Forecasting Using Deep Learning

Objective

The objective of this assignment was to evaluate different deep learning architectures for time-series weather forecasting and determine which approach produced the most accurate temperature predictions. Five neural network configurations were compared against a common-sense baseline using mean absolute error (MAE).

Methodology

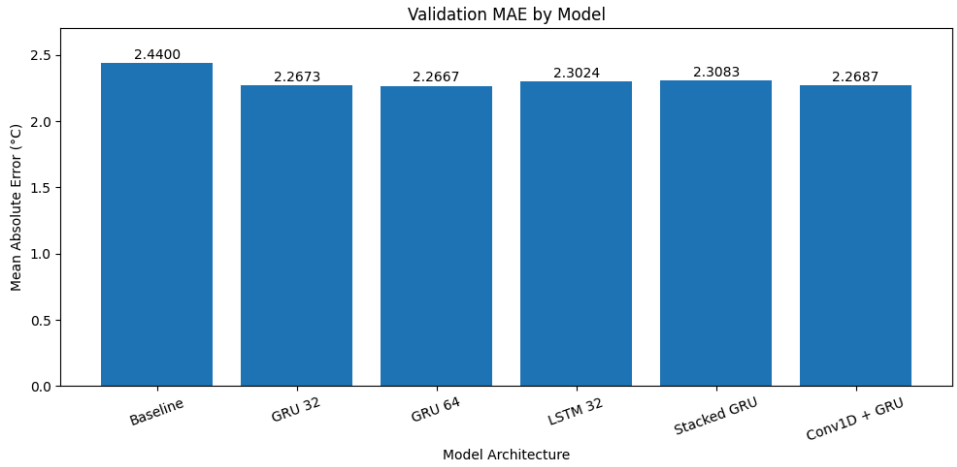
The Jena Climate dataset was used to predict temperature 24 hours into the future based on five days of historical weather measurements. The dataset was divided chronologically into 50% training, 25% validation, and 25% testing data. Weather measurements were normalized using statistics calculated from the training data.

Five models were evaluated: a 32-unit GRU, a 64-unit GRU, a 32-unit LSTM, a stacked GRU, and a combined Conv1D and GRU model. Dropout and early stopping were used to reduce overfitting. Models were compared using their lowest validation MAE, and the three strongest configurations were evaluated on the test dataset.

Results

Figure 1 compares the validation MAE across all five neural network configurations and the common-sense baseline. All five models achieved lower validation errors than the baseline of 2.4400°C.

Figure 1. Validation MAE by Model



The 64-unit GRU achieved the lowest validation MAE of 2.2667°C, closely followed by the 32-unit GRU at 2.2673°C and the Conv1D + GRU model at 2.2687°C.

The three strongest models were evaluated on the test dataset. The 64-unit GRU achieved the lowest test MAE of 2.4246°C, compared with 2.4362°C for the 32-unit GRU and 2.4633°C for the Conv1D + GRU model. The best model reduced test error by approximately 7.46% compared with the baseline of 2.6200°C.

Discussion

All five neural network configurations outperformed the common-sense baseline on validation data, demonstrating that deep learning can identify useful patterns in historical weather measurements.

Increasing the GRU from 32 to 64 units produced only a marginal improvement in validation accuracy. Similarly, adding a second GRU layer did not improve performance, suggesting that additional model complexity was not beneficial in these experiments.

The LSTM model performed slightly worse than both single-layer GRU configurations. The combined Conv1D and GRU architecture produced competitive validation results but had a higher test error than either single-layer GRU model.

Although the 64-unit GRU performed best, the differences among the strongest models were small. Additional training runs would help determine whether these differences are consistent.

Conclusion

The results demonstrate that recurrent neural networks can improve temperature forecasting compared with a simple baseline approach. The 64-unit GRU achieved the best overall performance, with a test MAE of 2.4246°C.

However, increasing architectural complexity did not consistently improve predictive accuracy. These findings suggest that selecting an appropriate model architecture and controlling overfitting are more important than simply increasing the number of layers or units.